

Community views on hydrogeology education: A global perspective from 2024

Stephanie L. Musy^{1*}, James M. Thornton², Ty Ferre³, Amanda Sills⁴, Daniel Hunkeler², Philip Brunner²

¹Hydrogeology, Environmental Science, University of Basel, Basel, Switzerland

²Center for Hydrogeology and Geothermics, University of Neuchâtel, Neuchâtel, Switzerland

³Hydrology and Atmospheric Sciences, University of Arizona, Tucson, United States

⁴G360 Institute for Groundwater Research, University of Guelph, Guelph, Canada

*Corresponding author: stephanie.musy@unibas.ch

Abstract

Groundwater is the Earth's largest reservoir of available freshwater. As climate change and increasing anthropogenic pressures continue to affect its availability and quality, the discipline of hydrogeology is evolving rapidly. Training enough professionals capable of understanding and managing groundwater resources effectively is therefore more critical than ever. However, there is a perceived global shortage of adequately trained hydrogeologists – raising concerns about society's capacity to address future groundwater-related challenges. To obtain quantitative data regarding this perception and to evaluate the current state of hydrogeology education and workforce dynamics more generally, a global survey was conducted of hydrogeology professionals, recent graduates, and current students. 573 individuals from 82 countries responded, providing contemporary insights into the discipline's demographics, education pathways, and perceived training gaps. Here, a synthesis of the key findings are presented – offering a snapshot of current trends and challenges in hydrogeology education from various perspectives. Notable findings include the observations that: (i) female hydrogeologists remain under-represented; (ii) some countries act as major “exporters” or “importers” of hydrogeologists; (iii) many practitioners consider new graduates to be insufficiently prepared; (iv) most hydrogeological work is conducted by those without formal education in the field; and (v) hydrogeology is expected to contribute to a growing range of challenges, calling for both disciplinary depth and interdisciplinary breadth in curricula. These findings highlight a need for better alignment between academic training and professional demands. The full dataset is openly shared, offering a resource for educators, researchers, and policymakers to support training and engagement with global water issues.

28

29 **Keywords**

- 30 • Hydrogeology education
- 31 • Groundwater management
- 32 • Groundwater science communication
- 33 • Groundwater and society
- 34 • Sustainability

35 **1 Introduction**

36 As pressures on groundwater resources intensify due to climate change, pollution, over-abstraction, and competing
37 land-use demands, hydrogeology has become increasingly critical to global water security and sustainability.
38 Those engaged in the discipline play a vital role in understanding, managing, and protecting groundwater systems,
39 and demand for hydrogeological expertise in both the public and private sectors has risen sharply (Gleeson et al.
40 2012), but without a corresponding increase in groundwater education opportunities (Cherry 2023). This, coupled
41 with anecdotal evidence of declining enrolment rates in hydrogeology degree courses, suggests that fewer
42 graduates are entering the field. This potential disconnect highlights the need to critically evaluate whether current
43 approaches to hydrogeology education are effectively communicating the relevance of the discipline and
44 equipping students with the knowledge, skills, and motivation needed to pursue related careers. Despite the field's
45 growing importance, there is a lack of understanding of how hydrogeology is taught globally, which training
46 pathways are most effective, and whether educational programs align with employers' needs and emerging
47 challenges. Note that in the following, education refers to formal courses conducted in a university or similar
48 setting. Training, meanwhile, refers to a broader range of activities, including short courses, summer schools, and
49 learning “on the job”.

50 Certain aspects of hydrogeological education – particularly the role of fieldwork and applied learning – have been
51 discussed extensively in the literature. For example, Fletcher (1994) described the implementation of field
52 laboratories that provide hands-on experience with instrumentation and data collection. Day-Lewis *et al.* (2006)
53 expanded this concept by developing an on-campus well field specifically designed for the education of
54 undergraduate hydrogeophysicists and hydrogeologists, enabling repeated access for instruction and research.
55 Exercises described by Lee (1998) and Singha (2008) integrated laboratory work with conceptual learning, while

McKay and Kammer (1999) demonstrated how hydrogeology can be effectively incorporated into mapping-based field camps. Innovative teaching tools like the low-cost “Darcy bottles” – simple experimental apparatus to demonstrate groundwater flow principles – proposed by Neupauer and Dennis (2010) offered additional entry points for practical learning. More recent contributions (e.g., Gleeson *et al.* 2012; Jimenez-Martinez 2023) emphasize field-integrated curricula that support the transition from conceptual understanding to site-specific modeling – an essential step in developing professional hydrogeological reasoning. Hakoun *et al.* (2013) further advocate for the close integration of classroom instruction and applied learning to strengthen both technical understanding and practical capacity.

Despite such previous explorations of the various teaching and learning approaches, there is still no consensus across universities, organizations (in the public, private, and third sectors), and students regarding which approach – or more likely combination of approaches – is most effective in building the hydrogeological understanding and practical capacity required across the discipline. In other words, to our knowledge, no prior study has triangulated the viewpoints of the three aforementioned primary stakeholder groups. Given the increasing responsibilities placed on hydrogeologists to address climate change, water scarcity, land-use pressures, and environmental degradation, this evidence gap must be urgently addressed.

One important consideration in addressing this gap is the prevalence of regional variations in hydrogeological systems and challenges (e.g., fractured rock, karst, Quaternary deposits, coastal aquifers), which shape the practical applications of hydrogeology and therefore market demand for related expertise. Consequently, the suitability of educational content and pedagogical approaches is typically highly context-dependent, making it difficult (if not impossible) to define a universally applicable set of core competencies for all hydrogeology graduates. Adding to this complexity is the current diversity in how hydrogeology is taught. Given the field’s hybrid nature – combining elements of geology, hydrology, and engineering – it is rarely offered as a standalone undergraduate degree. Instead, it is more commonly taught at the postgraduate level (e.g., MSc), or via professional certificates (e.g., Certificate of Advanced Studies (CAS) programs), short courses, or “on-the-job” learning.

However, no comprehensive global overview of these educational pathways currently exists – a deficiency that we aim to address here. A further challenge lies in the continual emergence of new topics with implications for groundwater (e.g., climate change, urbanization, and novel pollutants) and the rapid evolution of relevant technologies (e.g., numerical modeling, remote sensing, and artificial intelligence). These shifts are redefining the scope and expectations of hydrogeological practice. In light of these realities, we argue that decisions about what

and how to teach future hydrogeologists must be informed by a broad, diverse sample of the global professional community – rather than solely by small expert groups in specific regions. A concerted effort to re-examine and update long-standing definitions of the broad hydrogeology education is now overdue if we are to future-proof the discipline. We are confident that a better understanding of hydrogeologists’ demographics, training pathways, and career trajectories can support the development of more responsive and inclusive educational models.

In this context, the present paper offers a much-needed reality check and quantitative insights into pressing questions such as: Do students feel prepared for professional practice? Are universities teaching the skills that industry and other practice-oriented sectors require? And are educational innovations keeping pace with job market demands? To explore these issues, we present the core findings of a global survey conducted in conjunction with the International Association of Hydrogeologists (IAH) 2024 World Groundwater Congress (WGC) in Davos, Switzerland (IAH2024Davos 2024a). Hydrogeology education was a central focus of the Congress. The survey was launched ahead of the event and targeted a broad audience across the educational and professional spectrum, including students, academic instructors, and practicing hydrogeologists. Respondents were recruited through the WGC registration process, professional networks, academic institutions, and social media. At the WGC itself, hydrogeology education was further highlighted through a full-day workshop on teaching innovations and a plenary debate entitled “Future-Proofing Hydrogeology Education” (IAH2024Davos 2024b), which brought together diverse voices from academia, non-governmental organizations (NGOs), and industry to discuss key challenges and future directions.

The survey explored four main areas: (i) respondent demographics (e.g., location, professional role, and academic background); (ii) education and training pathways (e.g., degree programs, institutions, and field experiences); (iii) career trajectories (e.g., employment sectors, geographic mobility, and perceived preparedness); and (iv) perceived gaps in hydrogeology education, including both technical and conceptual skills, and its alignment with real-world needs. By capturing responses from across the educational and professional spectrum, the survey enables comparison of perspectives by role and region, offering a comprehensive view of the global hydrogeology education landscape.

This paper accompanies the publication of the full survey dataset (questions and results; Musy et al. 2025) that offers a timely snapshot of how hydrogeology education was perceived by the global community in 2024 and serves as a foundation for further research and discussion. We present selected core findings without aiming for exhaustive analysis or interpretation, as several other contributions in this special issue explore specific aspects of the dataset in greater depth. The **Key Survey outcomes** section outlines the current state of the hydrogeology

field. More specifically, the *Respondent demographics* section presents the demographics of survey participants, including gender, age, professional roles, and geographic distribution. The *Global supply of hydrogeology practitioners* section examines the perceived availability of trained hydrogeologists and the extent to which current educational systems are meeting professional demands. The *Current curricula* section discusses reported experiences and perceptions regarding hydrogeology education, highlighting areas considered well-covered and those viewed as needing improvement. The *Outlooks and future pathways of hydrogeology education* section explores future directions for hydrogeology education. The *Emerging challenges* section identifies expected areas of growth and decline within the disciplines. The *Promoting the field* section examines strategies to strengthen awareness and engagement. The *Potential solutions and ways forward* section outlines possible approaches to improve hydrogeology training. Finally, the *Conclusions* section summarizes the key findings and draws conclusions.

2 Key survey outcomes

2.1 Respondent demographics

In total, 573 individuals from 82 countries (spanning five continents) participated. **Fig. 1** shows the geographic distribution of respondents, with particularly high engagement from China, the United States, Switzerland, France, Germany, and the United Kingdom.

The survey captured a diverse cross-section of the hydrogeology community, including students, early-career professionals, and senior experts. The age distribution skewed younger, with over 35% of respondents being under 35 years old (**Fig. 2a**, Survey question No. 3), indicating strong engagement from early-career hydrogeologists. This trend is further supported by graduation year data: while respondents graduated between 1965 and 2024 (**Fig. 2b**), 55% completed their studies in the past 10 years. The sample was also highly educated, with 42% holding an MSc and 46% a PhD (question No. 5).

Most respondents with a dedicated post-graduate degree (i.e. MSc or PhD) in hydrogeology first obtained undergraduate degrees in Geology (50%), Environmental Science (12%), or Civil Engineering (8%) (question No. 10), emphasizing the specialized nature of hydrogeology programmes. In terms of professional roles, the majority of respondents reported working in academia (60%), followed by the private sector (39%). Only a small proportion were affiliated with NGOs (1%) or the public sector (0.5%) (question No. 18). This distribution likely reflects the academic context of the WGC, where the survey was conducted. Geographically, the respondents spanned all

inhabited continents, albeit with a clear skew towards the Global North. Europe (46%), Asia (20%), and North America (17%) were the most represented regions, likely reflecting the location of the WGC 2024 in Switzerland and the networks used to disseminate the survey.

Gender disparities were immediately apparent: 67% of respondents identified as male, 32% as female, and 1% chose not to specify (question No. 2). These imbalances are consistent with broader patterns observed across Science, Technology, Engineering, and Mathematics (STEM) fields, which are widely recognized for their persistent gender disparities (UNESCO 2020; Fru et al. 2021). Based on the survey results (**Fig. 2b**), only about one in four hydrogeology graduates in the 1990s were women. By the early 2000s, this proportion had risen to around 30% and has continued to increase modestly over the past 5–7 years. This trend indicates a gradual shift toward greater gender balance, though the field remains predominantly male-dominated. At present, men still make up around 65% of the respondents, with significant gender gaps persisting across all professional roles and academic levels (BSc, MSc, PhD; **Fig. 2c** and **d**). While gender diversity appears to be improving at entry-level education (i.e., BSc graduates), female hydrogeologists remain underrepresented in senior and leadership positions (question No. 2, **Fig. 2**). This imbalance, characterized by the over-representation of experienced men, has important implications for career development and educational culture. In particular, the lack of visible female role models hinders inclusion and long-term retention for younger women entering the field (Popp et al. 2019).

Further analysis of the results highlights significant disparities in gender representation across countries. A small number of countries such as Thailand, Mongolia, Slovenia, Portugal, Brazil, Poland, and Chile reported a hydrogeology workforce composed of more than 50% women. However, these figures should be interpreted with caution, as each of these countries had fewer than 10 respondents, making the results susceptible to sampling bias rather than indicative of broader national trends. In contrast, the majority of countries with larger response rates show a clear gender imbalance, with women representing a much smaller proportion of the workforce. This underrepresentation is particularly evident in countries such as Germany, India, the United States, Pakistan, and Ethiopia.

Fig. 3 considers the international mobility of hydrogeologists by illustrating each country's "net import ratio" – the ratio between the number of respondents working in a country and the number trained domestically, minus one. This ratio represents an approximation of mobility derived from survey responses and does not replace detailed institutional data on graduate destinations. A value of 0 indicates balance, meaning the country trains and retains enough hydrogeologists to meet its internal demands. Countries in near equilibrium include Brazil, the

United States, China, and Norway. Negative values (< 0) indicate net exporters of hydrogeologists, where more professionals are trained than remain in the country to work. This situation is evident in India, Canada, France, South Africa, Japan, Iran, and Germany, suggesting emigration of trained talent. Conversely, positive values (> 0) indicate net importers of hydrogeologists – that is, countries that rely on foreign-trained professionals, either to fill domestic workforce gaps or, in some cases, because foreign professionals outcompete locally trained ones (e.g., through stronger qualifications or lower wage demands). Notable examples in this category include Zambia, Peru, Kenya, New Zealand, Australia, Switzerland, and Austria. Note that in this context, it is important to consider that a large proportion of the respondents (60%) were from academia, where international mobility is common.

2.2 Global supply of hydrogeology practitioners

Fig. 4 reveals a widespread consensus across all types of professional affiliations and career stages that there is a shortage of adequately trained hydrogeologists seeking to enter the profession. When responses are disaggregated by continent (**Fig. S1** of the electronic supplementary material (ESM)), this perception is evident across all regions represented in the survey. However, its apparent intensity varies geographically, with respondents from Oceania and Europe reporting the highest proportions of perceived shortage. These regional patterns should be interpreted with caution, as they may be influenced by the uneven geographical distribution of survey participants. Other results suggest that this gap is even more pronounced at the mid-to-senior level (i.e., experienced practitioners are especially scarce; see question No. 23).

This shortage of trained hydrogeology professionals translates into relatively long hiring processes, with approximately 40% of recruitments taking more than four months regardless of the seniority level of the open position (question No. 24). According to the survey responses, several factors may explain this persistent shortage besides the general drop in enrollment in Earth Science-related programs (Bonaccorsi et al. 2020; Streetly 2023; Anderson 2023; Moss et al. 2025). One key issue is the lack of awareness among STEM undergraduates about career pathways in hydrogeology, which may lead to fewer students pursuing a career in the discipline. This is symptomatic of the very low degree of awareness about the origin of domestic water in general populations, especially in developed countries (Seelen et al. 2019; Brouwer et al. 2021). At the same time, demand for hydrogeologists has increased and is expected to continue rising – e.g. by an estimated 3–4% in the United States between 2024 and 2034 (US Bureau of Labor Statistics 2025), largely in response to climate change and growing anthropogenic pressures on groundwater resources. Education and training opportunities, however, have not kept pace with this demand (Cherry 2023; Ferre 2024), contributing to a persistent mismatch between market needs

and academic preparation. Survey responses also highlighted that limited career progression opportunities might further discourage professionals from entering the field, while competition from other disciplines offering more attractive career prospects and better working conditions could draw potential hydrogeologists elsewhere. Another major factor is that hydrogeologists' salaries are typically relatively low, either in absolute terms or compared to other STEM fields, making the discipline a less attractive career choice (question No. 25).

Interestingly, the survey results indicate that only about 10% of hydrogeology practitioners hold a degree specifically in hydrogeology (**Fig. 5a**). When trained hydrogeologists are unavailable, the private and government sectors hire professionals with different academic backgrounds or rely on consultants. In most cases, the individuals employed in professional hydrogeology-related roles instead hold geology, environmental sciences, or civil engineering degrees (Fig. 5b), reflecting a widespread reliance on adjacent disciplines to fill posts.

Since hydrogeology lies at the intersection of multiple disciplines, with groundwater systems closely linked to the atmosphere, lithosphere, and biosphere, the diverse profiles of individuals working in the discipline are in some ways unsurprising, and such interconnections and contributions from other scientific fields can undoubtedly be extremely valuable in addressing hydrogeological challenges. For example, the prevalence of professionals with backgrounds in geology or environmental science brings important strengths in understanding the associated geologic and ecological systems, respectively. As groundwater investigations increasingly rely on quantitative tools and system-based modeling, any gaps in specialized training can become a liability for robust groundwater system characterization, prediction, and decision-making. While highly specialized or site-specific expertise can be developed through professional practice rather than university education, a solid grounding in hydrogeological principles, quantitative reasoning, and conceptual modelling remains important for informed decision-making. In complex hydrogeological settings, such as volcanic or karstic aquifers, limited familiarity with groundwater-specific concepts may increase uncertainty in assessments of flow, storage, and vulnerability. Because the consequences of groundwater-related decisions often emerge only over long timescales, responsibility is typically shared across institutions, employers, and practitioners rather than attributable to individual actors. In this context, maintaining clear, discipline-specific competencies, while fostering interdisciplinary collaboration, is essential to ensure that hydrogeology continues to serve societal needs effectively.

2.3 Current curricula

In an ideal scenario, there should be alignment between the perceptions of early-career hydrogeologists regarding their education and how employers evaluate the preparedness of newly graduated hires. However, the survey

results reveal a significant mismatch between these perspectives among respondents working in the private sector, public sector, and NGOs (**Fig. 6**). While 65% of recent graduates (defined here as respondents who began working within the last three years) felt their studies had prepared them extremely well or rather well for a career in hydrogeology, 62% of employers viewed entry-level hires as moderately to severely under-trained. This discrepancy underscores a clear gap between academic training (or at least the extent to which graduates grasp it) and industry expectations, pointing to the need for better alignment between curricula and workforce demands. Importantly, as discussed earlier, many professionals hired into hydrogeology roles lack formal training in the field – an issue that may contribute to employers’ concerns about preparedness.

When respondents were asked which topics are insufficiently covered in current hydrogeology programs, the top three identified areas were coastal hydrogeology, groundwater modeling, and groundwater resource development or management (question No. 9). While regional factors may influence these responses, the prominence of these topics highlights their growing significance in addressing climate change-related challenges. In parallel, the topics considered essential for hydrogeology programs by respondents include sustainability of groundwater resources, groundwater–surface water interactions, water quality and pollution, advanced data acquisition, and hydrogeological engineering (question No. 16). These priorities reflect a growing emphasis on sustainable groundwater management, which is becoming increasingly critical in the face of population growth and elevated anthropogenic pressures on water resources.

Responses to questions on teaching modes (question Nos. 30–33) indicate a clear preference for in-person tuition across all sub-areas—fundamental theory, field skills, computer skills, and soft skills. While hybrid formats were also rated relatively highly, fully online modes (particularly autonomous ones) were generally considered less effective, especially for field techniques. This pattern holds across age groups, although younger respondents tend to be slightly less adverse to online approaches (**Fig. S2** of the ESM). This is notable given the growing trend toward online delivery in higher education, suggesting that hydrogeology’s practical and applied nature still benefits strongly from face-to-face learning. Regarding software training (question No. 44), MODFLOW emerged as the most frequently prioritized package, reflecting its open-source status and central role in groundwater modelling. Python was also highly valued, likely due to its versatility in data analysis and modelling, as well as in GIS tools such as ArcGIS and QGIS. Responses to question No. 45 suggest a pragmatic approach to software choice—most respondents favour a “mostly open-source” model but support the inclusion of commercial software where it offers clear advantages.

3 Outlooks and future pathways of hydrogeology education

3.1 Emerging challenges

Fig. 7 highlights projected shifts in the importance of different hydrogeology-related fields. Some fields, such as climate change adaptation, groundwater protection, and groundwater production for drinking water supply, are expected to grow significantly in the coming years (Fig. 7), aligning with emerging climatic challenges and sustainability objectives. These results closely align with the topics that were identified as being insufficiently covered in current hydrogeology curricula, as well as those deemed essential for training programs, in the *Current curricula* section. Conversely, fields linked to industrial development, such as mining, industrial applications, and civil engineering-related hydrogeology, are expected to decline (Fig. 7). This may indicate a collective expectation of a gradual shift away from extractive industries toward more environmentally conscious practices. However, some caution is warranted since the projection may reflect demographic or regional biases within the survey population, many of whom work in academic or public-sector roles based in high-income countries. In other regions, especially where industrial expansion continues, these fields may remain central to hydrogeological work for many years to come.

Importantly, the differences in the frequency with which respondents identified potential growth areas were relatively modest, with support ranging from 3% to 14% across the 13 predefined categories. This suggests that there is no singular or dominant vision for the future of hydrogeology. Instead, the responses point to a diversification of the field, with many respondents anticipating growth across a broad spectrum of applications. The fact that every proposed area received some level of support reflects a general belief that hydrogeology will evolve along multiple, parallel tracks, rather than converging towards a narrow specialization. Interestingly, some topics appeared in both the growth and decline categories, suggesting that regional differences in hydrogeological work, priorities, and economic contexts significantly shape perceptions of relevance and future demand.

This diversity highlights the need for educational frameworks that, while providing solid core knowledge, are also both broad and adaptive, capable of preparing students for a range of emerging challenges. It underscores the growing importance of interdisciplinary and environmentally focused training, while also reinforcing the value of curricular flexibility to accommodate regional and sector-specific demands. Preparing future hydrogeologists will thus require programs that balance technical depth with the capacity to engage with complex, systems-level issues across a wide spectrum of applications.

3.2 Promoting the field

As supported by the survey data (question No. 25), the shortage of graduate hydrogeologists entering the profession may be partly underpinned by the general lack of awareness among the public about the importance of groundwater and its role in water supply systems. This underlying gap in knowledge can influence several of the more specific factors identified in the survey, such as limited awareness of career pathways, the relative attractiveness of other fields, and the availability of dedicated university courses. To address this, the hydrogeology community must strengthen communication and outreach efforts. More specifically, there is a need for greater public engagement and more effective science communication, including increased media presence and accessible educational content to help demystify hydrogeology for a broader audience. Better early-age education is equally crucial, with stronger advocacy for integrating fundamental water-related concepts – such as the water cycle – into junior school curricula. The Groundwater Project, for example, offers freely available books aimed at a wide readership, including specific categories for Children’s Education and Introductory Books, which serve as excellent tools for raising awareness from an early age (gw-project). Hydrogeologists could also make greater efforts to share their research findings in formats accessible to young people, such as those offered by the journal *Frontiers: Young Minds* (e.g. Pearson and Aitchison-Earl 2022).

The relatively low participation of female hydrologists could be addressed by the establishment of international and/or intersectoral female–female mentoring schemes, and female–only field camps. Initiatives like “Girls on Ice”, which combine field science and mentorship for young women and further demonstrate how targeted outreach can promote inclusivity while fostering early interest in geosciences, could provide inspiration (Inspiring Girls International 2022). Policymakers should be encouraged to prioritize water education, recognizing its essential role in sustainability and resource management.

On a more technical level, hydrogeology as a discipline needs to better establish its unique advantages compared to, and interrelationship with, other adjacent fields. If undertaken, this should contribute to individuals who have obtained dedicated hydrogeology qualifications being responsible for a greater proportion of professional hydrogeological activities in the future.

3.3 Potential solutions and ways forward

Multiple viable pathways exist to meet the evolving educational needs of the hydrogeology profession and thereby begin to overcome some of the challenges identified in this paper. Reflecting the fact that different students have different needs (not only as a function of region but also economic status, other responsibilities, time availability

etc.), these include fully online programs, traditional in-person university-based degrees, short-term professional courses, and many hybrid models that combine online instruction with in-person field or lab components (e.g., SYMPLE 2021). Each approach offers distinct advantages depending on regional context, institutional resources, and learner needs. Given that a significant number of hydrogeology professionals do not have a background in hydrogeology, specialized short-term professional courses will undoubtedly remain of critical importance.

As alluded to already, in designing future courses, it will likely become increasingly necessary to take account of the profound changes in technology and (to some extent) data acquisition which are relevant to hydrogeology – whilst many of the applications of hydrogeology remain well known, given advances in computer power, artificial intelligence, remote sensing, and numerical modelling, the approaches that hydrogeologists will adopt to conduct their work are likely to continue to evolve considerably. Courses should therefore arguably not only cover classical theory and current technology but also convey a capacity for adaptability and technological prowess amongst graduates. This will prove especially useful in the light of the fact that another responsibility of hydrogeology educators and curricula is to produce individuals capable of going on to conduct advanced research (i.e. PhD and beyond) related to groundwater.

Limiting the tendency for there to be a “brain drain” of trained hydrogeologists from developing countries to areas with higher salaries and better infrastructure will require efforts to be made to enhance the opportunities, conditions, and support for hydrogeologists to pursue a career within their home countries (or the countries in which they were trained, if different), for instance through internationally financed positions or projects related to climate change adaptation. The success of such initiatives will stand and fall with the perceived importance of groundwater on a governmental level. To improve alignment between graduand / early career hydrologists’ perceptions of their preparedness for practice on the one hand, and employers on the other, it could be useful to explicitly involve or consult various industry and public-sector stakeholders in the development of curricula (if not already done). Voluntary or mandatory practical placements or projects, potentially even involving international exchanges to gain wider perspectives, as integral course components, could also be pursued.

Debate around the most effective methods for training future hydrogeologists will undoubtedly continue. The various sessions and short courses held at the 2024 IAH WGC in Davos that were focused on hydrogeological education culminated in a plenary panel discussion that addressed current shortcomings, persistent challenges, and emerging opportunities in the field. These conversations reflected a diversity of perspectives, from academic educators to industry practitioners. Importantly, the congress also catalyzed a wave of more detailed contributions

that build upon our global survey, including other articles on specific topics in this issue. Together, these efforts represent steps towards more structured reflection on the future of hydrogeological training. The openly available dataset generated through the survey provides a useful starting point for ongoing dialogue. While the sample may not fully represent the global hydrogeology community (see the *Respondent demographics* section), we are confident that it offers rich insight into the demographics, priorities, and concerns of professionals engaged with the field across five continents.

Efforts made before, during, and after the WGC are already bearing fruit, for instance by shaping discussions within institutions, guiding curriculum revisions, and inspiring new research directions. At the same time, much work remains to be done. The survey results are not an endpoint, but a shared resource for educators, students, researchers, and practitioners to reflect on where the field is heading, and how best to prepare for its complex and evolving future.

4 Conclusions

As global demand for water intensifies, hydrogeology faces mounting challenges driven by climate change, over-abstraction, pollution, and growing anthropogenic pressures. Addressing these issues requires a well-trained workforce capable of understanding complex subsurface processes and managing groundwater sustainably – particularly in vulnerable or rapidly evolving environments. However, the survey results introduced here highlight a general shortage of adequately trained hydrogeologists. This shortage appears linked to low public awareness of groundwater, limited visibility of the profession, and reduced appeal compared to other scientific careers. Significant gaps in current curricula were also identified, particularly in areas of emerging importance such as climate change adaptation, groundwater quality protection, and long-term sustainability. We hope that the results of this survey stimulate further reflection on how the visibility of hydrogeology can be strengthened and encourage more targeted discussions and initiatives within the community on pathways to promote the field more effectively. These findings point not only to a need for curricular evolution but also to the importance of clarifying and better aligning expectations regarding the respective roles of universities and employers in professional training. While academic programs are well placed to provide strong conceptual foundations and analytical skills, some applied competencies are developed through professional practice. Improved dialogue between universities and employers may therefore help ensure that educational outcomes and workforce expectations are better matched.

Bridging the gap between academic training and professional needs will require closer collaboration among universities, industry, and public-sector water authorities. Priorities include expanding hands-on training,

incorporating real-world case studies into coursework, and building partnerships that support job readiness. Additionally, persistent gender disparities – particularly in senior roles – point to the need for reforms that promote equity, mentorship, and inclusion throughout career pathways.

Despite current challenges, the strong engagement of early-career professionals in the survey offers hope for the future. Their perspectives can guide needed reforms in education and career development strategies. It is important to acknowledge that the survey sample is not fully representative of the global hydrogeological community. Academics and PhD holders from the Global North are strongly overrepresented, while practitioners in consulting, government, or non-academic roles – especially those without doctoral training – are underrepresented. Rather than a limitation, this highlights the need for expanded future outreach that includes a wider range of voices and experiences across regions and sectors.

Encouragingly, new online and university-led digital learning initiatives are expanding access to hydrogeology education. A proposal for a common curriculum has also emerged, aiming to standardize core training and improve professional mobility. However, any such framework must remain flexible to support shared standards while allowing programs to adapt to regional realities and local challenges. Ultimately, securing the future of hydrogeology will require sustained investment in education, inclusive workforce development, and greater public and political support. The profession must continue to evolve – through broader engagement, innovative partnerships, and continued dialogue – to meet the many urgent groundwater-related challenges of the decades ahead. We hope this study contributes to that process by helping align academic training with the realities of professional practice.

394 **Acknowledgment**

395 The authors would like to sincerely thank all 573 individuals who participated in the survey. Your time, insights,
396 and contributions were invaluable to this study and to advancing the conversation around the future of
397 hydrogeology education. We also thank the two reviewers, Dylan Irvine and Luke Peeters, as well as the guest
398 editor, Ilka Wallis, for their thoughtful and constructive comments, which helped improve the clarity and balance
399 of the manuscript.

400 **Conflicts of Interest**

401 The authors declare that they have no conflicts of interest relevant to the content of this article.

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462

Declarations

Ethics approval and consent to participate

Ethical approval was not required for this study as it involved voluntary participation from adult professionals and students, anonymised, and non-sensitive data. Participants were informed about the purpose of the study, and consent was implied through survey completion.

Consent for publication

All participants were informed about the purpose of the survey and that anonymised, aggregated data would be used for research publication. Consent was implied through completion of the survey.

Availability of data and material

The anonymized datasets generated during and/or analysed during the current study are available in the HydroShare repository: Musy, S. L., J. M. Thornton, T. Ferre, A. Sills, D. Hunkeler, J. Cherry, P. Brunner (2025). Community views on hydrogeology education: Global Survey results from the 2024 International Association of Hydrogeologists' Meeting, HydroShare, <https://doi.org/10.4211/hs.4beea6712df548dea264619a14d6441d>. The survey questionnaire and a README file describing data handling, anonymization procedures, and respondent privacy protections are also available.

Competing interests

The authors have no conflicts of interest to declare that are relevant to the content of this article.

Funding

The authors did not receive support from any organization for the submitted work.

Author contributions

All authors contributed to the study conception and design. The survey preparation and data collection were performed by Stephanie L. Musy, James M. Thornton, Ty Ferre, Amanda Sills, Daniel Hunkeler, and Philip Brunner. The data analysis was done, and the first draft of the manuscript was written by Stephanie L. Musy, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

FIGURE CAPTIONS

Fig. 1: Geographic distribution of survey respondents by country of employment. The map uses the Equal Earth Greenwich coordinate reference system (EPSG:8857).

Fig. 2: Gender distribution of hydrogeology practitioners across the (a) age categories, (b) graduation years, (c) professional roles, and (d) highest degree obtained. Column and point graphics represent the absolute counts (left axis). In (b), the smooth line represents the trend in gender percentage over graduation years.

Fig. 3: Net importation of hydrogeology practitioners. The values represent the ratio between the number of hydrogeologists working in a country and the number trained domestically, minus one. A value of 0 indicates a net balance; Values below 0 indicate net emigration of domestically trained professionals; Values above 0 reflect net immigration.

Fig. 4: Distribution of responses to the question “Is there a shortage or oversupply of graduate hydrogeologists seeking to enter the profession?” (Survey question No. 21)

Fig. 5: Distribution of academic disciplines among hydrogeology professionals. (a) Fields of study for respondents’ highest qualifications (Survey question No. 11), and (b) Alternative disciplines of hired professionals when no trained hydrogeologist is available (question No. 27).

Fig. 6: Perspectives regarding the preparedness of freshly-graduated hydrogeologists: (left) professional hydrogeologists who began working within the last three years, reflecting on how well their education prepared them for their roles (Survey question No. 20); and (right) practicing hydrogeologists who have recently recruited

514 entry-level staff (i.e., in the last three years), assessing the theoretical knowledge and practical skills of new hires
515 upon starting to work (question No. 22).

516

517 **Fig. 7:** (left) Major current or likely future decline areas in hydrogeology; (right) major areas of potential growth
518 in hydrogeology, according to the survey respondents (Survey question No. 34-35). For both questions,
519 respondents could select multiple options from 13 predefined categories and had the opportunity to provide
520 additional input through an open-text field.

521

Figure 1

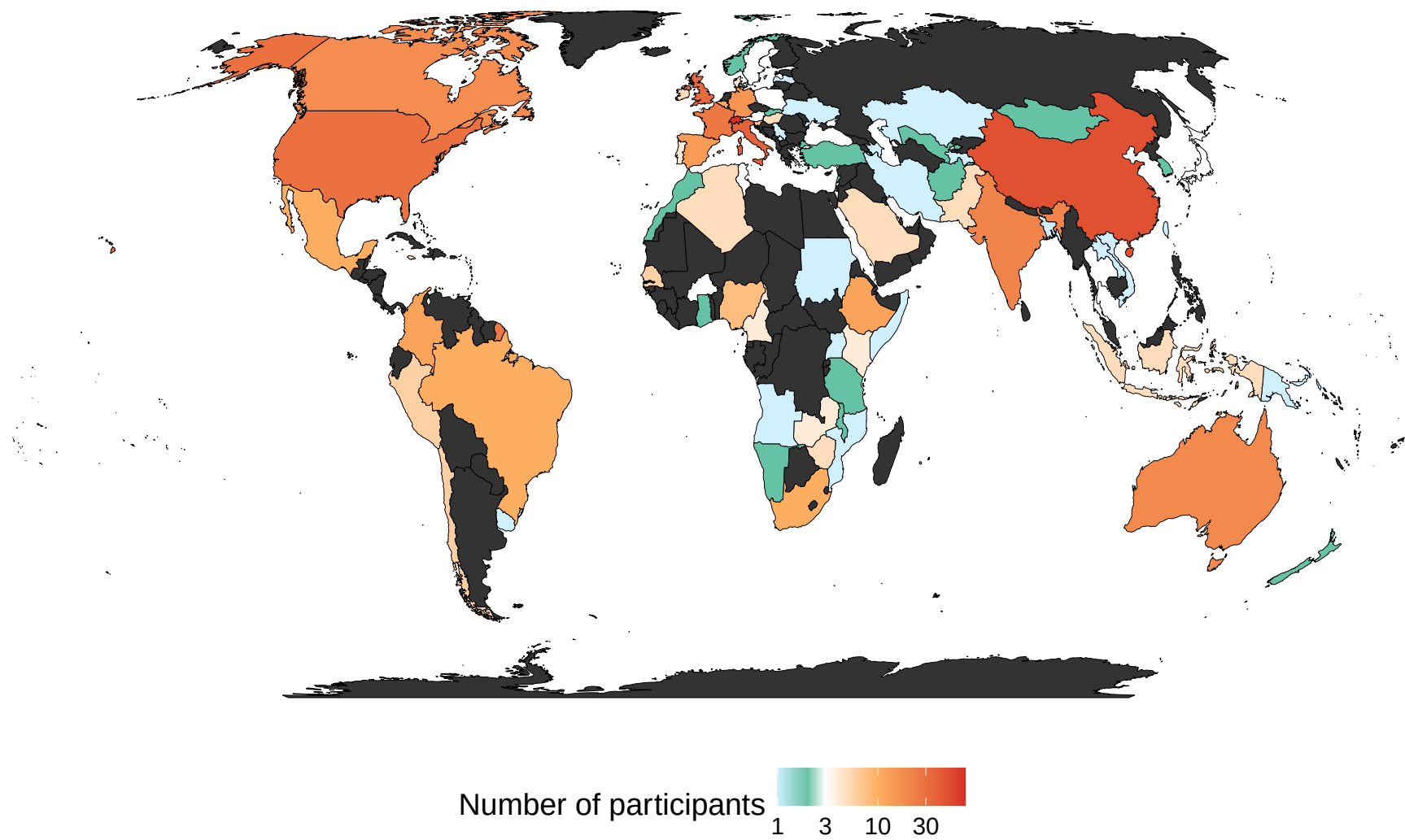
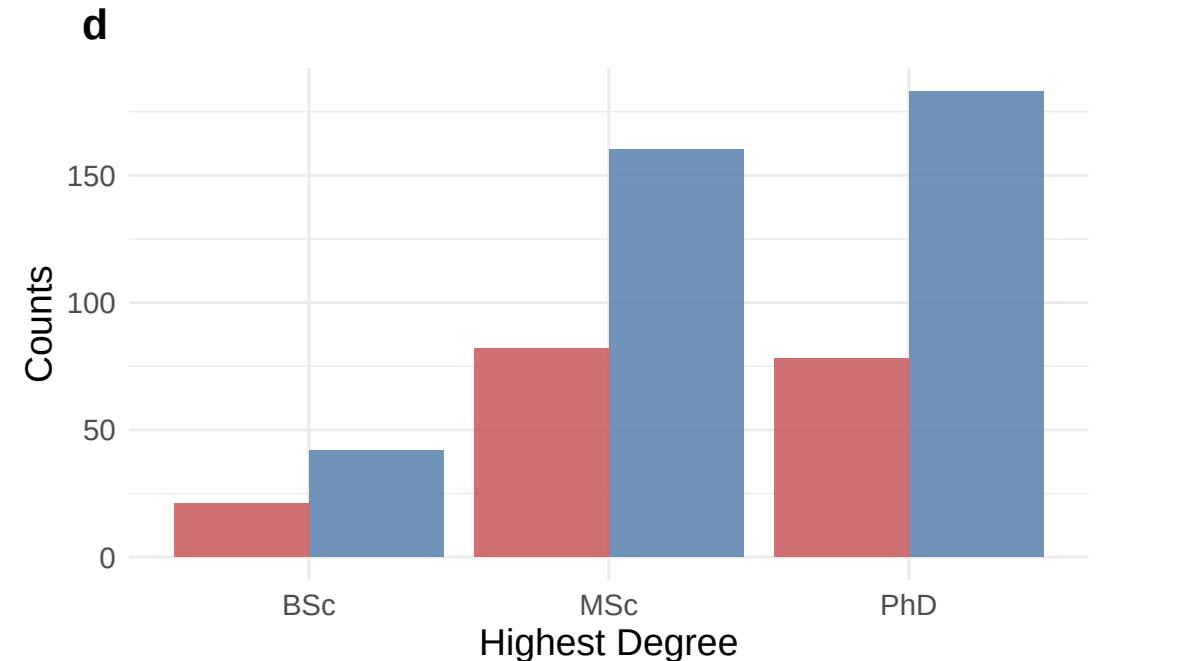
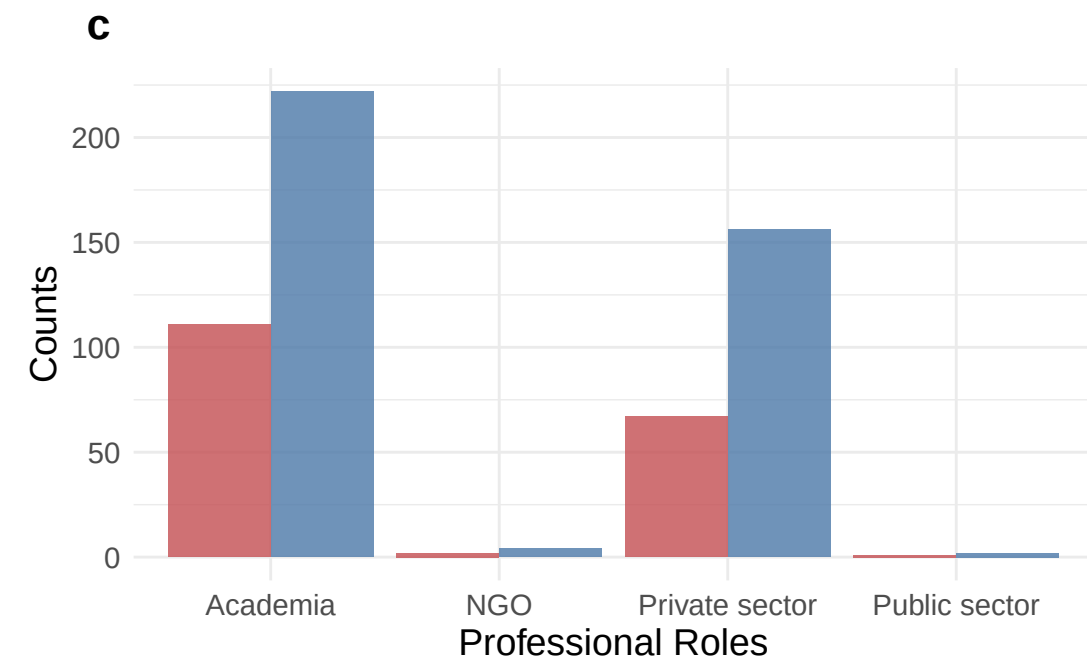
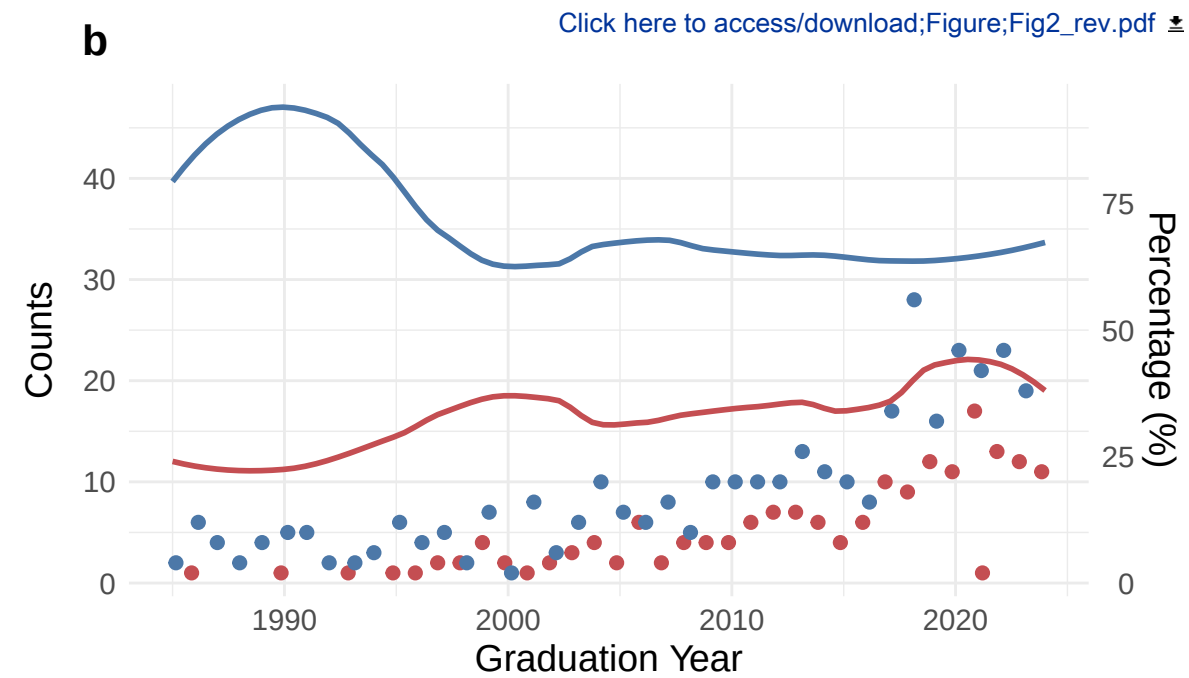
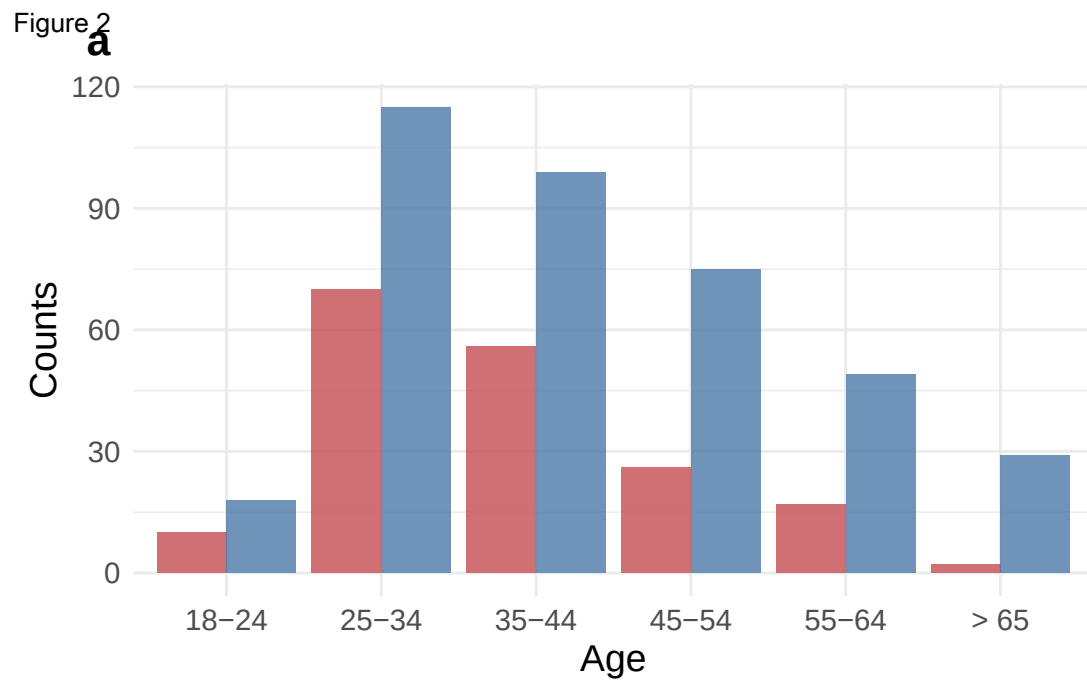


Figure 2

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Gender Female Male

Figure 3

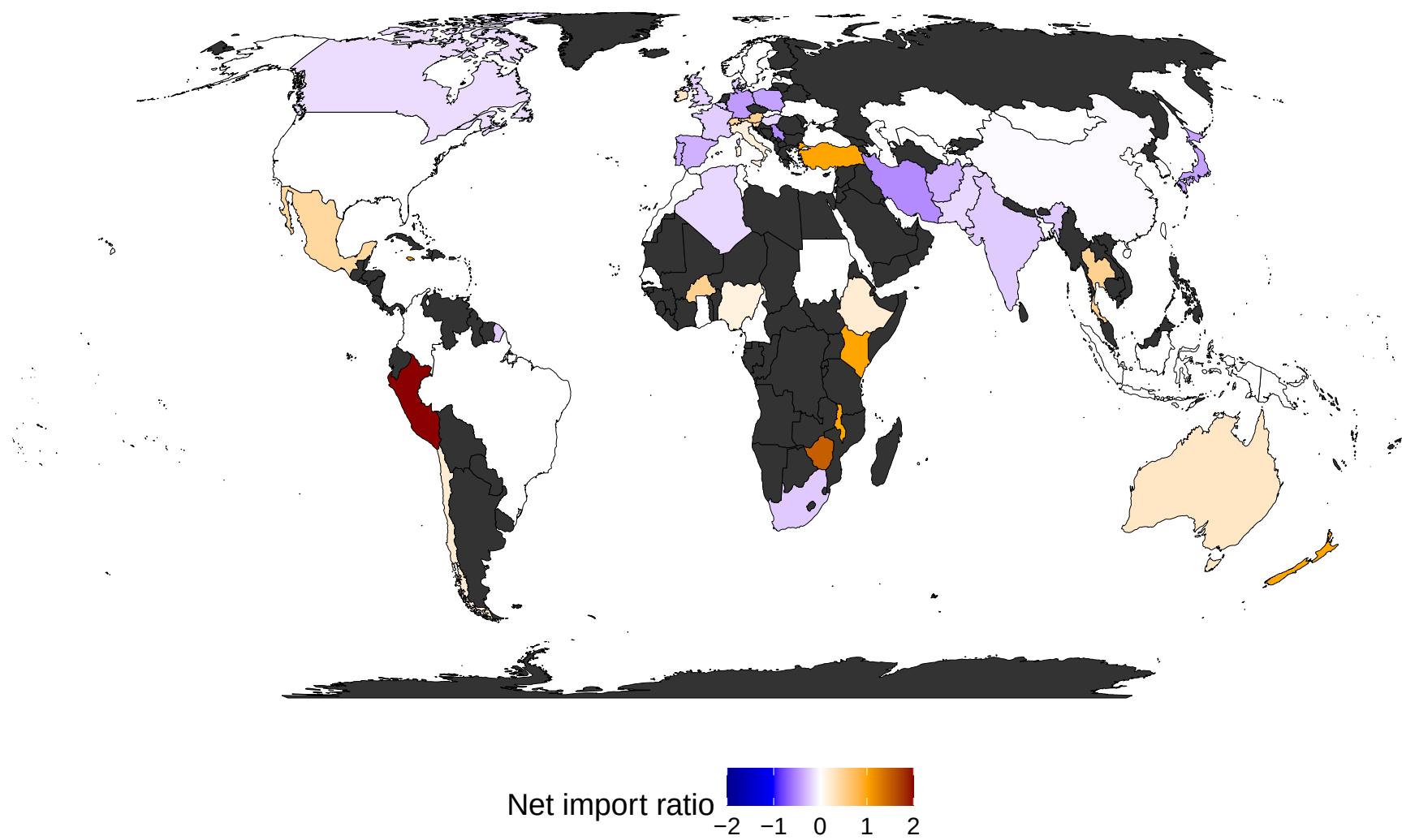
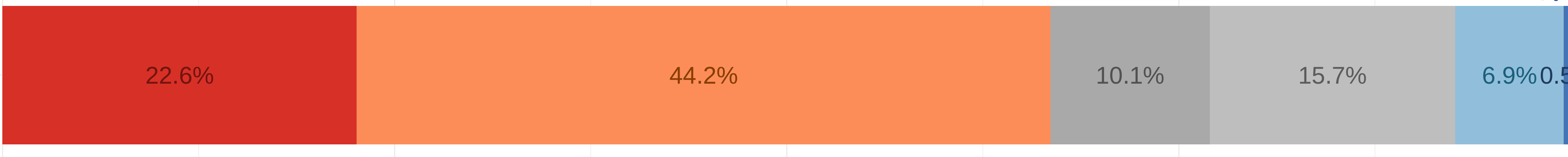


Figure 4

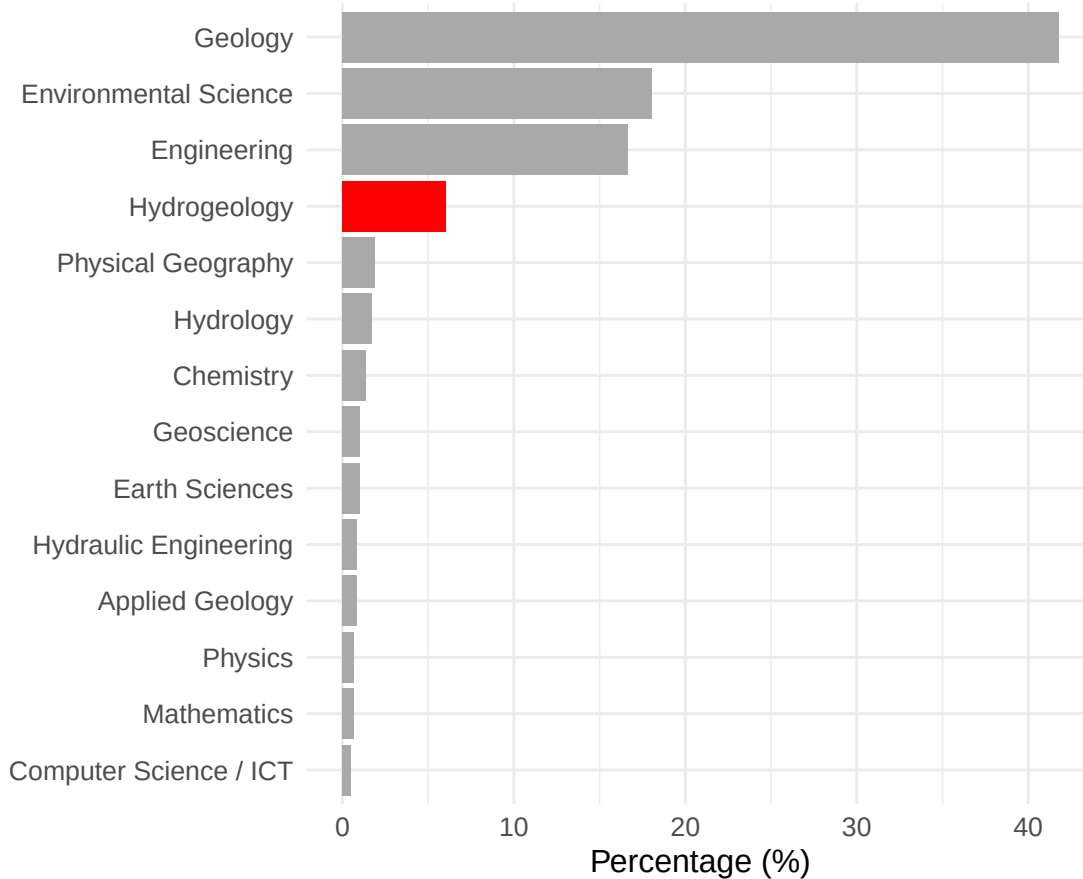
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■ Severe shortage ■ Moderate shortage ■ Not sure ■ Neither shortage nor oversupply ■ Moderate oversupply ■ Severe oversupply

Figure 5

a



b

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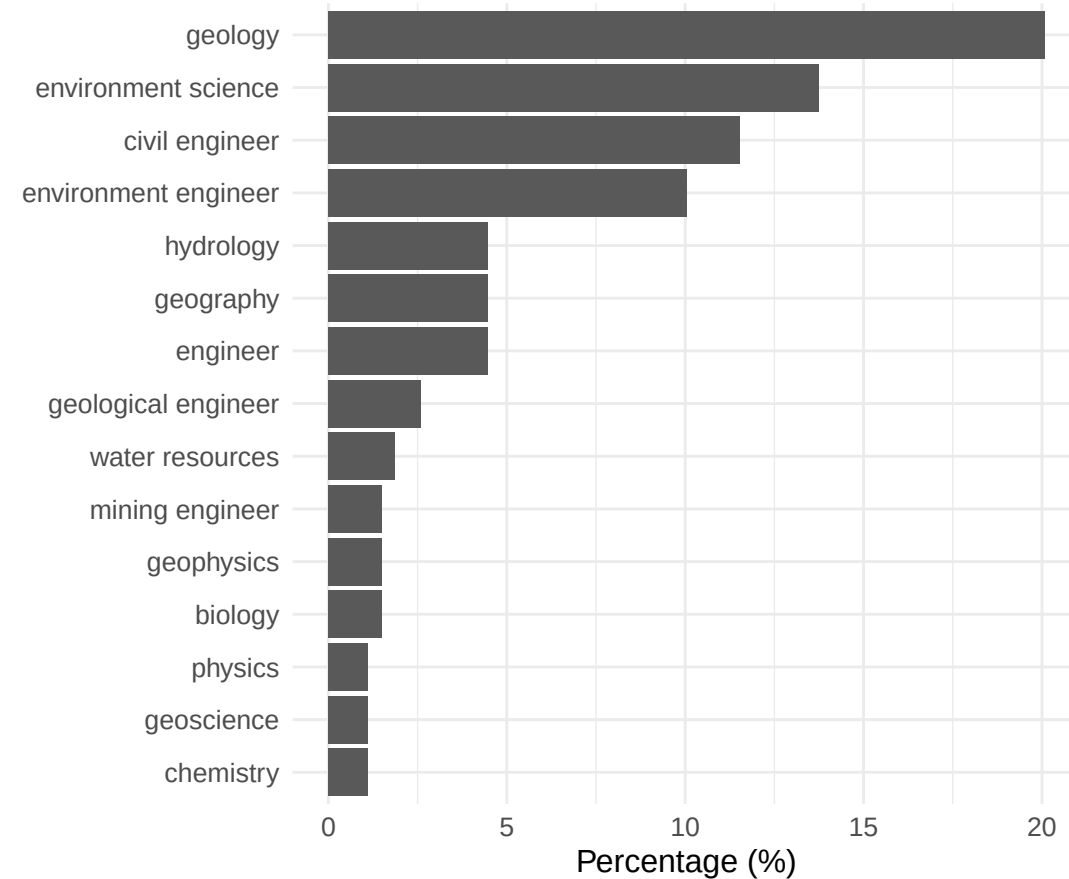
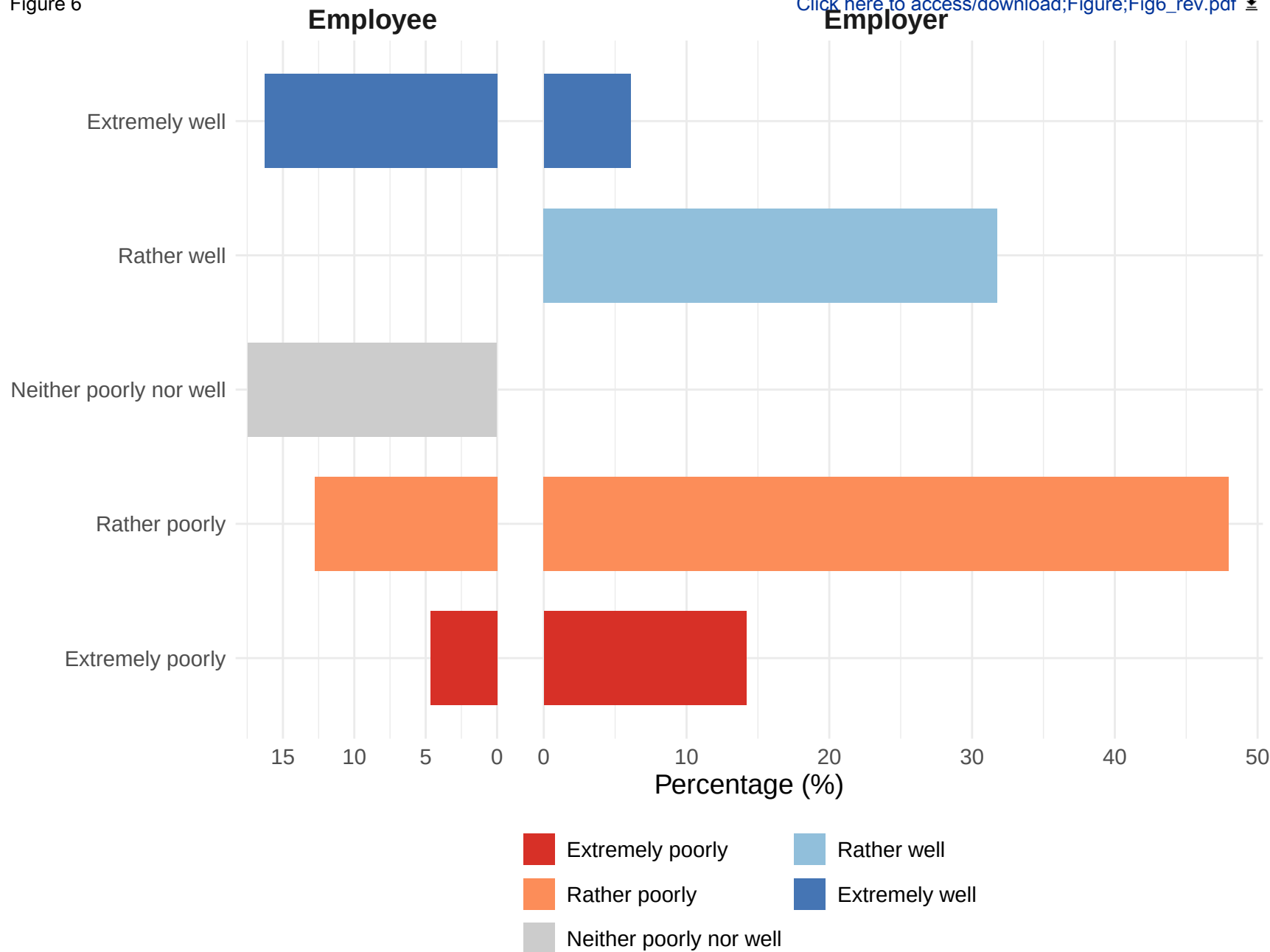


Figure 6

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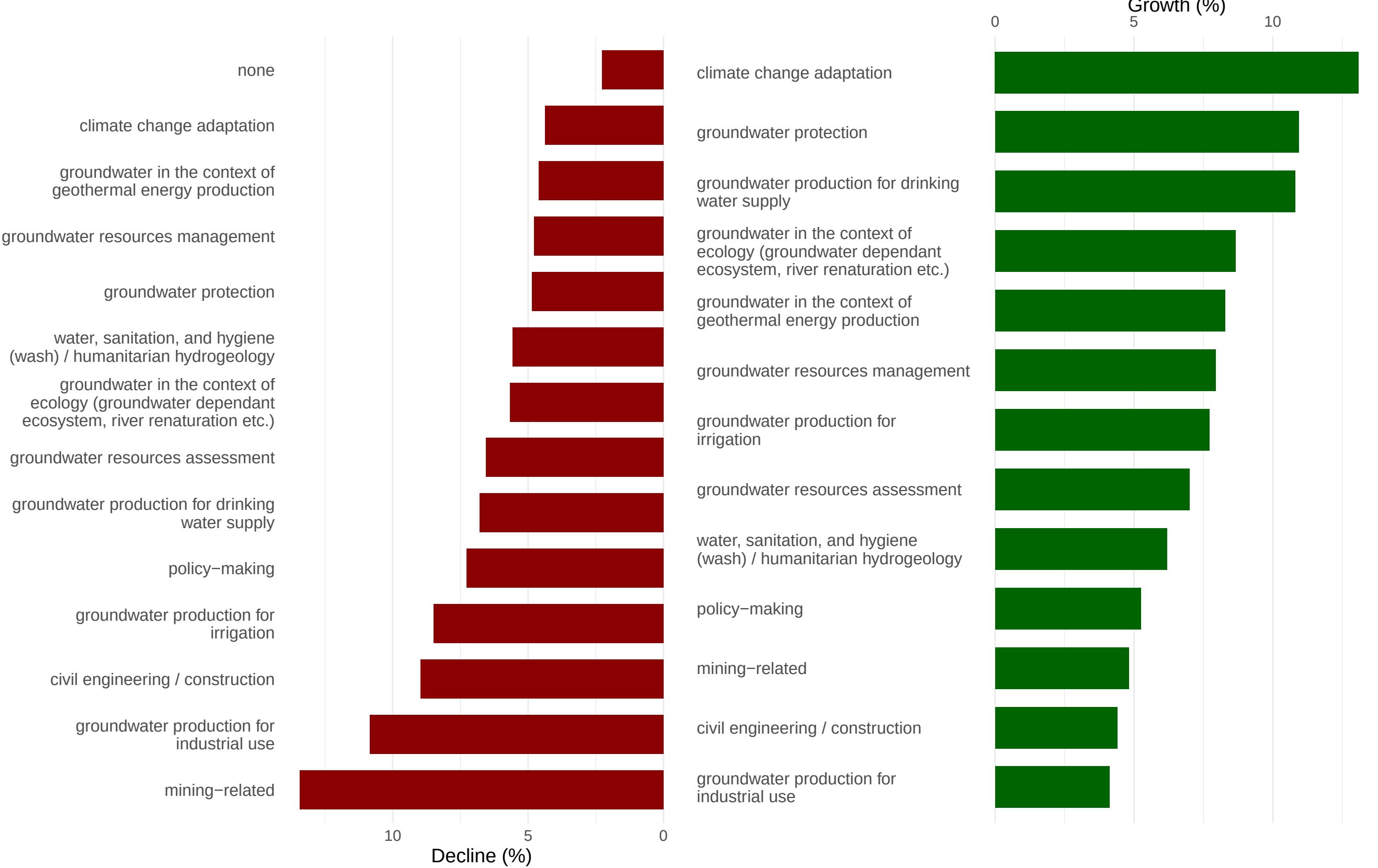


Figure S1

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